

Data sources and risks

DATA SCIENCE FOR BUSINESS



Michael Chow
Data Scientist

Common sources of data

- Web events
- Customer data
- Logistics data
- Financial transactions

Web data

- Events
- Timestamps
- User information

user_id	event_name	timestamp
1234	homepage_visit	2019-01-01 12:01:01

Personally Identifiable Information (PII)

Name	Timestamp	Object Clicked
Jane Doe	2019-01-20 12:05:00	Like Button

"Jane Doe" = Personally Identifiable Information (PII)

Data pseudonymization

user_id	Timestamp	Object Clicked
185477	2019-01-20 12:05:00	Like Button

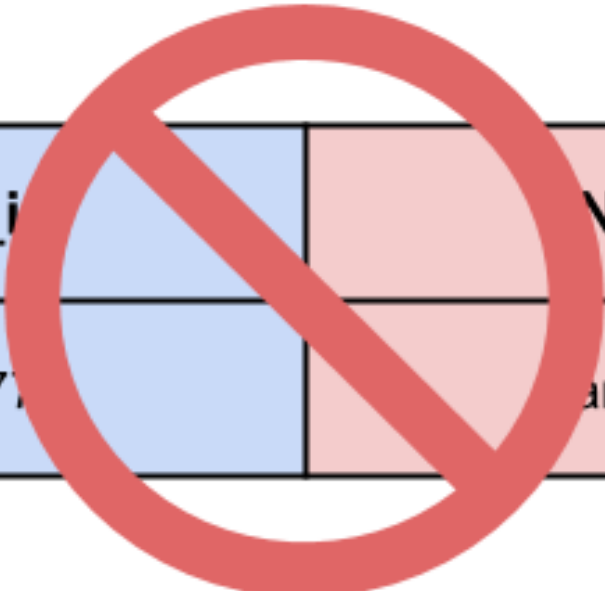
user_id	Name
185477	Jane Doe

- Restricted access
- Audit logs

Data anonymization

user_id	Timestamp	Object Clicked
185477	2019-01-20 12:05:00	Like Button

user_id	Name
185477	Jane Doe



General Data Protection Regulation (GDPR)

- Applies to all data inside of the EU
- Give individuals control over their personal data
- Regulates how long data can be stored
- Mandates appropriate anonymization
- Disclose data collection and gain consent

Let's practice!

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Solicited data

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Why do we solicit data?

- Create marketing collateral
- De-risk decision making
- Monitor quality



Types of solicited data

- Surveys
- Customer reviews
- In-app questionnaires
- Focus groups

We appreciate your feedback!

X

Thank you for visiting our website. We are always looking for ways to improve your experience. Please take a moment to tell us about your experience.

How likely are you to recommend our website to a friend or colleague?

0

1

2

3

4

5

6

7

8

9

10

What could we do to improve your experience?

Send Feedback

powered by

 QuestionPro

Types of solicited data

Qualitative

- Conversations
- Open-ended questions

Quantitative

- Multiple choice
- Rating scale

Revealed and stated preferences

Stated preference

- Hypothetical
- Subjective



Revealed preference

- Actions
- Purchasing decisions



Best practices

Be specific

Do this	Not that
On a scale of 1 - 5, how would you rate the quality of content on DataCamp?	How would you rate DataCamp?

Best practices

Be specific

Do this	Not that
On a scale of 1 - 5, how would you rate the quality of content on DataCamp?	How would you rate DataCamp?

Avoid loaded language

Do this	Not that
Which of the following political issues is most important to you?	Which of the following controversial political issues is most important to you?

Best practices

Calibrate

Do this	Not that
Rate your interest in each of the following products at DataCamp.	Are you interested in Skill Assessment at DataCamp?

Best practices

Calibrate

Do this	Not that
Rate your interest in each of the following products at DataCamp.	Are you interested in Skill Assessment at DataCamp?

Require actionable results

Do this	Not that
Have a hypothesis for each question.	Ask a question just because it's interesting.

Let's practice!

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Collecting additional data

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Even more data

- APIs
- Public records
- Mechanical Turk



Data APIs

- Application Programming Interface
- Request data over the internet
- Twitter
- Wikipedia
- Yahoo! Finance
- Google Maps
- Many more!

Tracking a hashtag

- All tweets with `#DataFramed` (DataCamp's podcast!)
- Use Twitter API

DataFramed

DataCamp's official podcast. Presented by [Hugo Bowne-Anderson](#).

Data Science is one of the fastest growing industries and has been called the « Sexiest job of the 21st Century ». But what exactly is Data Science? In the podcast by DataCamp, Hugo Bowne-Anderson approaches this question from the perspective of what problems Data Science tries to solve instead of what definition fits it best. From automated medical diagnosis and self-driving cars to recommendation systems and climate change, come on a journey with industry and academic experts to explore the inner workings of the industry that will color the 21st century.

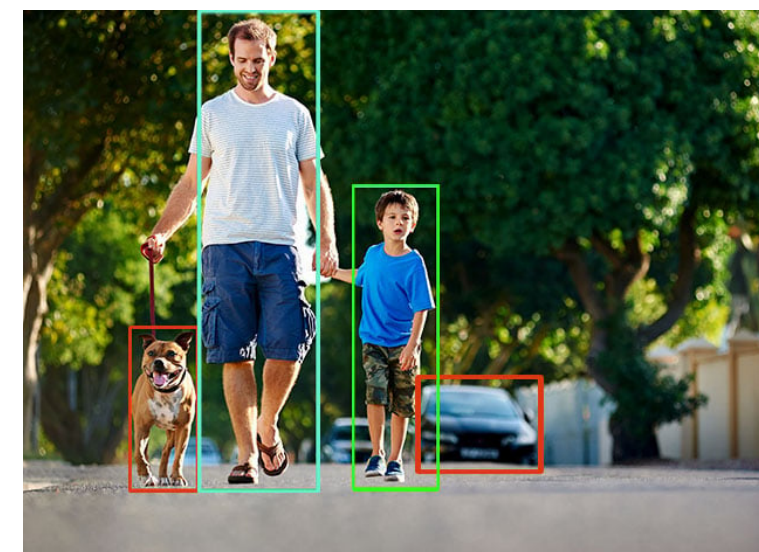
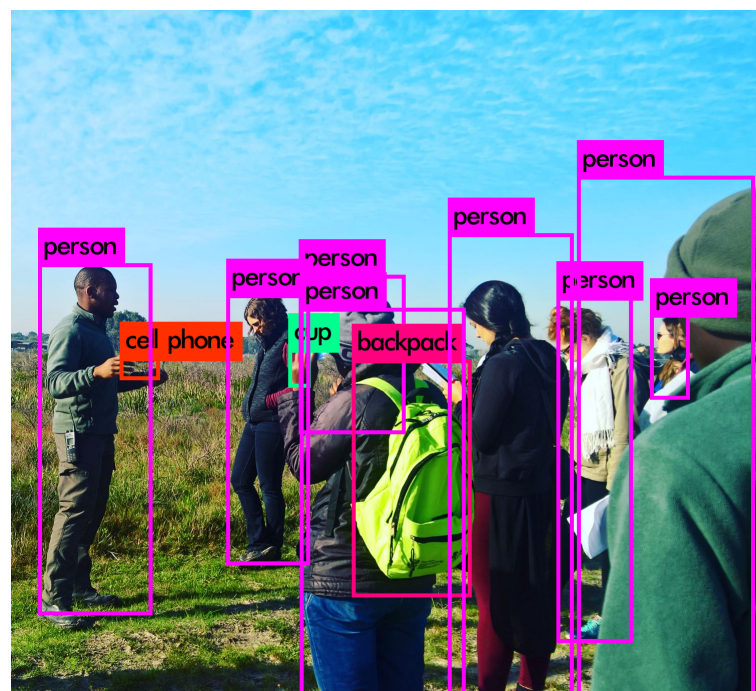
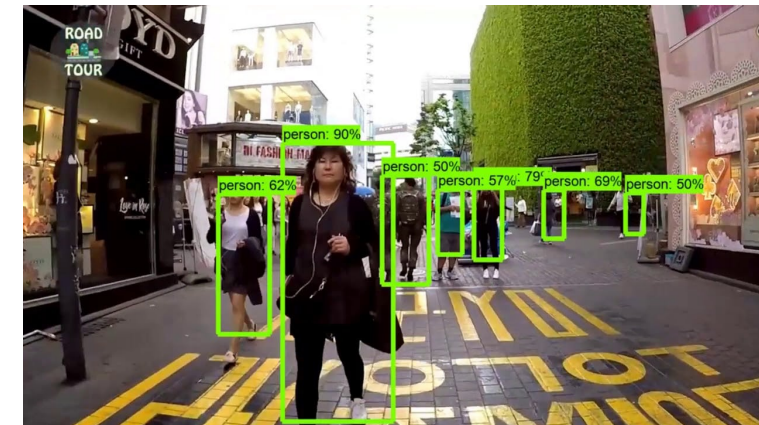
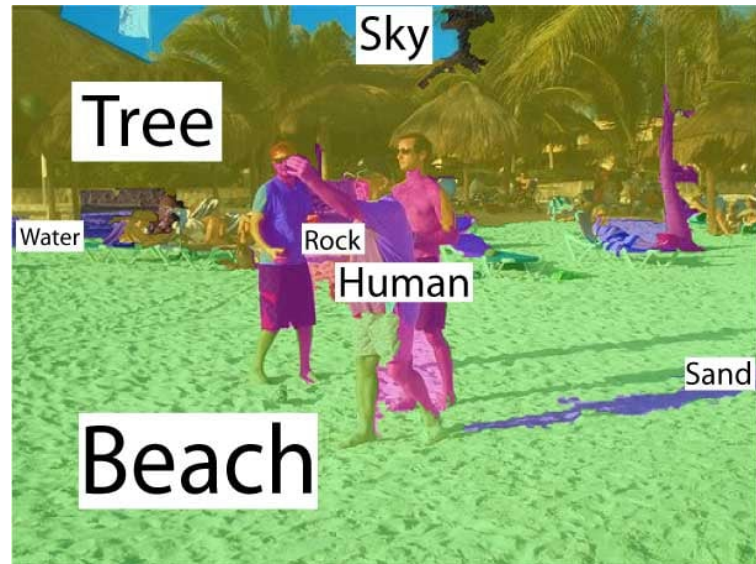


Public records

- For the US, data.gov
- For the EU, data.europa.eu

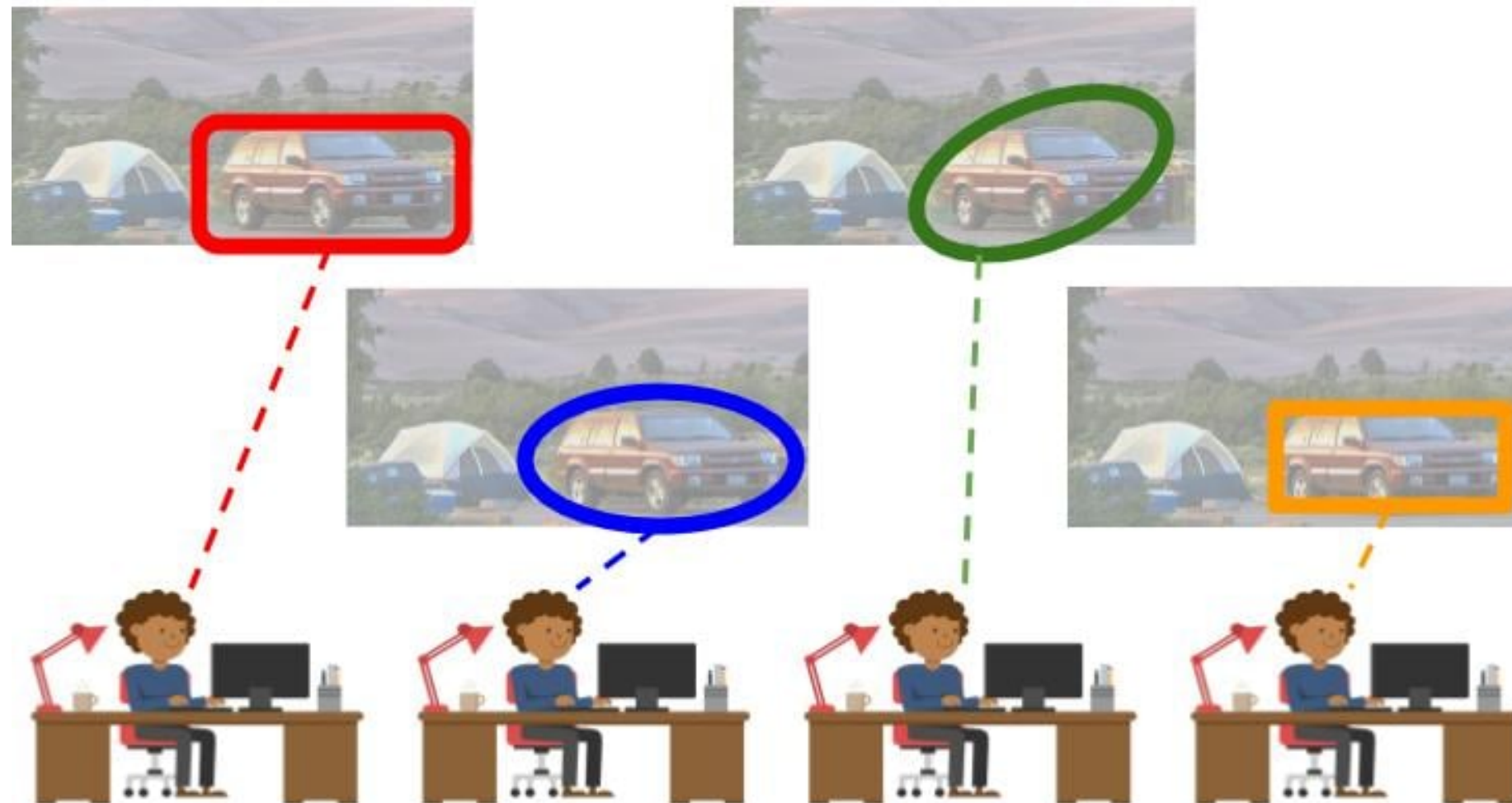


Building a training set



Mechanical Turk

Select the car in the image.



Mechanical Turk

- Resource: AWS MTurk
- Label customer reviews
- Extract text from a form
- Highlight key words in a sentence

Jane
Last Name
Smith
Email
stopall
Pick your color
☒ Red
☐ Green
☐
Submit

Select all squares with street signs.



⌂ 🎧 ⓘ

VERIFY

Let's practice!

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Data storage and retrieval

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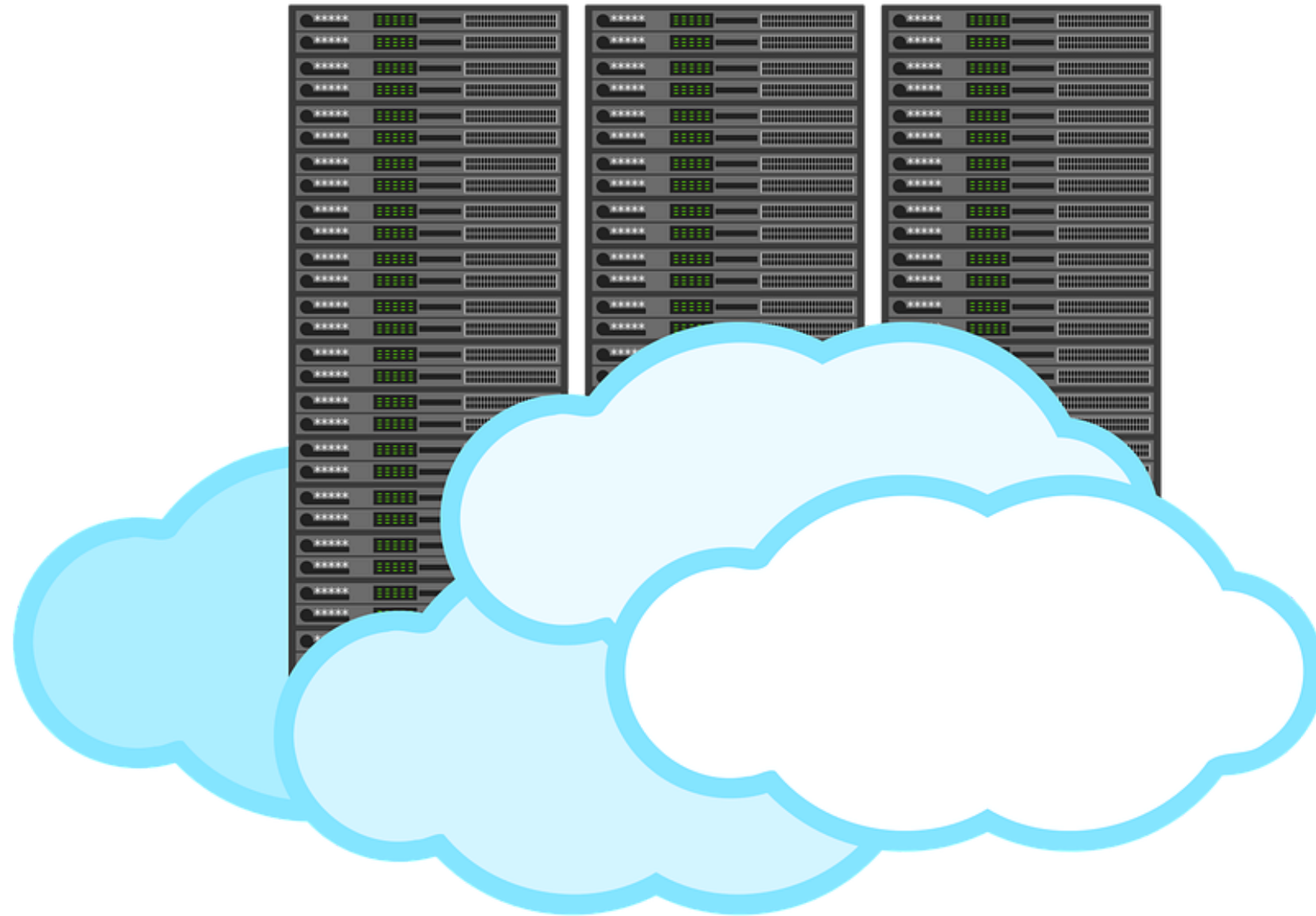


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Parallel storage solutions



The cloud



Types of data storage

Unstructured

- Email
- Text
- Video and audio files
- Web pages
- Social media

Document Database

Types of data storage

Unstructured

- Email
- Text
- Video and audio files
- Web pages
- Social media

Document Database

Tabular

Customer Name	Customer Address	...
Jane Doe	123 Maple St.	...

Relational Database

Data querying



Data querying



Data Type	Query Language
Document Database	NoSQL
Relational Database	SQL

Putting it all together: Location



- On-premises cluster
- Cloud provider:
 - Azure
 - AWS
 - Google Cloud

Putting it all together: Data type



Putting it all together: Data type

Data Type	Storage Solution
Unstructured	Document Database
Tabular	Relational Database



Putting it all together: Queries



Putting it all together: Queries



Data Type	Query Language
Document Database	NoSQL
Relational Database	SQL

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